



Top 4 CAT DI Miscellaneous Questions With Video Solutions

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Questions

Instructions

Each visitor to an amusement park needs to buy a ticket. Tickets can be Platinum, Gold, or Economy. Visitors are classified as Old, Middle-aged, or Young. The following facts are known about visitors and ticket sales on a particular day:

1. 140 tickets were sold.
2. The number of Middle-aged visitors was twice the number of Old visitors, while the number of Young visitors was twice the number of Middle-aged visitors.
3. Young visitors bought 38 of the 55 Economy tickets that were sold, and they bought half the total number of Platinum tickets that were sold.
4. The number of Gold tickets bought by Old visitors was equal to the number of Economy tickets bought by Old visitors.

Question 1

If the number of Old visitors buying Gold tickets was strictly greater than the number of Young visitors buying Gold tickets, then the number of Middle-aged visitors buying Gold tickets was

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4. The number of Gold tickets bought by Old visitors was equal to the number of Economy tickets bought by Old visitors.

	P	G	E
O			
M			
Y			

140

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Question 2

If the number of Old visitors buying Platinum tickets was equal to the number of Middle-aged visitors buying Economy tickets, then the number of Old visitors buying Gold tickets was

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	P	G	E
O			
M			
Y			

140

VIDEO SOLUTION

Question 3

If the number of Old visitors buying Platinum tickets was equal to the number of Middle-aged visitors buying Platinum tickets, then which among the following could be the total number of Platinum tickets sold?

- A 34
- B 36
- C 38
- D 32

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	P	G	E
O			
M			
Y			

140

VIDEO SOLUTION

CAT Mock Test

Mocks designed exactly like the actual CAT

MOCK

Question 4

Which of the following statements MUST be FALSE?

- A The numbers of Gold and Platinum tickets bought by Young visitors were equal
- B The numbers of Middle-aged and Young visitors buying Gold tickets were equal
- C The numbers of Old and Middle-aged visitors buying Platinum tickets were equal
- D The numbers of Old and Middle-aged visitors buying Economy tickets were equal

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	P	G	E
O			
M			
Y			

140

VIDEO SOLUTION

Answers

1.0

2.3

3.D

4.D

Explanations

1.0

Let 'x' be the the number of Old visitors. Then, the number of middle-aged visitors = $2x$.

Also, the number of Young visitors = $2 \cdot 2x = 4x$

$$\Rightarrow x + 2x + 4x = 140$$

$$\Rightarrow x = 20$$

It is given that total of 55 Economy tickets were sold out.

It is given that Young visitors half the total number of Platinum tickets that were sold.

Let 'Y' be the number of Platinum tickets bought by the Young visitors.

Then, the number of Platinum tickets sold = $2Y$.

Consequently, we can say that the number of Gold tickets sold = $140 - 55 - 2Y = 85 - 2Y$.

	Platinum	Gold	Economy	Total
Old				20
Middle-aged				40
Young	Y		38	80
Total	2Y	85-2Y	55	140

Let us assume that 'Z' is the number of Economy tickets bought by the Old visitors. It is given that the number of Gold tickets bought by Old visitors was equal to the number of Economy tickets bought by Old visitors.

	Platinum	Gold	Economy	Total
Old	$20 - 2Z$	Z	Z	20
Middle-aged	$(Y + 2Z) - 20$	$43 - (Y + Z)$	$17 - Z$	40
Young	Y	$42 - Y$	38	80
Total	2Y	85-2Y	55	140

It is given that the number of Old visitors buying Gold tickets was strictly greater than the number of Young visitors buying Gold tickets.

$$Z > 42 - Y$$

$$\Rightarrow Z + Y > 42 \dots (1)$$

The number of Middle-aged visitors buying Gold tickets = $43 - (Y + Z)$

Since $(Y + Z) > 42$, then We can say that $(Y + Z)_{\min} = 43$.

Hence, the number of Middle-aged visitors buying Gold tickets = $43 - 43 = 0$

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2.3

Let 'x' be the the number of Old visitors. Then, the number of middle-aged visitors = $2x$.

Also, the number of Young visitors = $2 \cdot 2x = 4x$

$$\Rightarrow x + 2x + 4x = 140$$

$$\Rightarrow x = 20$$

It is given that total of 55 Economy tickets were sold out.

It is given that Young visitors half the total number of Platinum tickets that were sold.

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Young	Y	$42 - Y$	38	80
Total	2Y	85-2Y	55	140

It is given that the number of Old visitors buying Platinum tickets was equal to the number of Middle-aged visitors buying Economy tickets.

$$20 - 2Z = 17 - Z$$

$$\Rightarrow Z = 3$$

Therefore, we can say that the number of Old visitors buying Gold tickets = 3

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3. D

Let 'x' be the the number of Old visitors. Then, the number of middle-aged visitors = $2x$.

Also, the number of Young visitors = $2 \times 2x = 4x$

$$\Rightarrow x + 2x + 4x = 140$$

$$\Rightarrow x = 20$$

It is given that total of 55 Economy tickets were sold out.

It is given that Young visitors half the total number of Platinum tickets that were sold.

Let 'Y' be the number of Platinum tickets bought by the Young visitors.

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	Platinum	Gold	Economy	Total
Old	20 - 2Z	Z	Z	20
Middle-aged	(Y+2Z) - 20	43 - (Y+Z)	17 - Z	40
Young	Y	42 - Y	38	80
Total	2Y	85-2Y	55	140

It is given that the number of Old visitors buying Platinum tickets was equal to the number of Middle-aged visitors buying Platinum tickets.

$$20 - 2Z = (Y+2Z) - 20$$

$$Y + 4Z = 40$$

$$2Y + 8Z = 80$$

$$2Y = 80 - 8Z$$

We can see that Z can take only integer values. Therefore, we can say that the the total number of Platinum tickets sold will be a multiple of 8. Hence, option D is the correct answer.

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4. D

Let 'x' be the the number of Old visitors. Then, the number of middle-aged visitors = 2x.

Also, the number of Young visitors = 2*2x = 4x

$$\Rightarrow x+2x+4x = 140$$

$$\Rightarrow x = 20$$

It is given that total of 55 Economy tickets were sold out.

It is given that Young visitors half the total number of Platinum tickets that were sold.

Let 'Y' be the number of Platinum tickets bought by the Young visitors.

Then,the number of Platinum tickets sold = 2Y.

Consequently, we can say that the number of Gold tickets sold = 140 - 55 - 2Y = 85 - 2Y.

	Platinum	Gold	Economy	Total
Old				20
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Total	2Y	85-2Y	55	140

Let us assume that 'Z' is the number of Economy tickets bought by the Old visitors. It is given that the number of Gold tickets bought by Old visitors was equal to the number of Economy tickets bought by Old visitors.

	Platinum	Gold	Economy	Total
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Total	2Y	85-2Y	55	140

Let us check with the help of options.

Option (A): The numbers of Gold and Platinum tickets bought by Young visitors were equal.

$$Y = 42 - Y$$

$$\Rightarrow Y = 21. \text{ Hence, this statement can be true.}$$

Option (B): The numbers of Middle-aged and Young visitors buying Gold tickets were equal

$$43 - (Y+Z) = 42 - Y$$

$\Rightarrow Z = 1$. Hence, this statement can be true.

Option (C): The numbers of Old and Middle-aged visitors buying Platinum tickets were equal

$$20 - 2Z = (Y + 2Z) - 20$$

$\Rightarrow Y + 4Z = 40$. Hence, this statement can be true.

Option (D): The numbers of Old and Middle-aged visitors buying Economy tickets were equal

$$Z = 17 - Z$$

$\Rightarrow Z = 8.5$. This is not possible as Z has to be an integer. Hence, we can say that this statement is false.

 VIDEO SOLUTION

